Differential Geometry

Shing Tak Lam

October 5, 2023

Lecture 1

1 Manifolds

The concept of a manifold is intended to generalise curves and surfaces in $\mathbb{R}^3$.

Definition 1.1 (smooth structure)
A *smooth structure* on a topological space M is a collection of *charts* (an *atlas*)

$$\varphi_\alpha : U_\alpha \to V_\alpha$$

where $U_\alpha \subseteq M$, $V_\alpha \subseteq \mathbb{R}^d$ are open, φ_α is a homeomorphism, such that

1. $M = \bigcup_\alpha U_\alpha$. That is, the charts cover M.
2. Given any charts $\varphi_\alpha, \varphi_\beta$, the map

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

   is a smooth map between open subsets of $\mathbb{R}^d$.
3. If φ is *compatible* (as in the previous point) with every chart in the atlas, then φ is a chart in the atlas.

In practice (i.e. using Zorn's lemma), given any collection of charts satisfying the first two conditions, we can find a maximal atlas containing it.

Remark 1.2.
1. The second condition implies that $\varphi_\beta \circ \varphi_\alpha^{-1}$ is a diffeomorphism.
2. d is fixed, assuming that M is connected.

Definition 1.3 (manifold)
A *(smooth) manifold* M is a second countable Huasdorff space topological space, equipped with a smooth structure.
$d = \dim(M)$ is the *dimension* of M.

In practice, we may induce a topology on M from a smooth structure (without the mention of open or homeomorphism), then $D \subseteq M$ is open if for any chart φ_α, $\varphi_\alpha(D \cap U_\alpha)$ is an open subseteq of $\mathbb{R}^d$.

Remark 1.4.
1. We can also define a C^k manifold by replacing smooth with C^k as above. If $k = 0$, then we have a *topological manifold*.
2. On the other hand, we can replace $\mathbb{R}^d$ with $\mathbb{C}^n$, and obtain a *complex manifold*.

$\mathbb{R}^n$ is (trivially) a manifold.

Example 1.5 (unit sphere)

Define

$$S^n = \{x = (x_0, \ldots, x_n) \in \mathbb{R}^n \mid ||x|| = 1\} \subseteq \mathbb{R}^{n+1}$$

For the charts, we can consider the stereographic projections

$$\varphi(x) = \frac{1}{1 - x_0}(x_1, \ldots, x_n) \qquad \text{for } x_0 \neq 1$$

and

$$\psi(x) = \frac{1}{1 + x_0}(x_1, \ldots, x_n) \qquad \text{for } x_0 \neq -1$$

In particular,

$$\psi \circ \varphi^{-1}(u) = \frac{u}{||u||^2}$$

Example 1.6

If M, N are manifolds of dimensions d, e respectively, then $M \times N$ is a manifold of dimension $d + e$.

As a corollary, $T^n = (S^1)^n$ is a manifold.

Example 1.7

If M is a manifold, $X \subseteq M$ is open, then X is a manifold.